Pandas Introduction (Detailed Guide for PDF)

What is Pandas?

Pandas is a powerful Python library used for data manipulation and data analysis. It is built on top of NumPy and provides easy-to-use data structures.

Why Use Pandas?

- Easy data handling
- Works with large datasets
- Powerful data analysis tools
- Used in data science and machine learning

Key Data Structures

Series

One-dimensional labeled array

```
import pandas as pd
s = pd.Series([1, 2, 3])
```

DataFrame

Two-dimensional table

```
data = {
  'Name': ['John', 'Alice'],
  'Age': [25, 30]
}

df = pd.DataFrame(data)
```

Installing Pandas

```
pip install pandas
```

Importing Pandas

```
import pandas as pd
```

Reading Data

```
df = pd.read_csv('file.csv')
df = pd.read_excel('file.xlsx')
```

Viewing Data

```
df.head()
df.tail()
df.info()
df.describe()
```

Selecting Data

```
df['Name']
df[['Name','Age']]
df.iloc[0]
```

Filtering Data

```
df[df['Age'] > 25]
```

Adding Columns

```
df['Salary'] = [50000, 60000]
```

Updating Data

```
df.loc[0, 'Age'] = 26
```

Deleting Data

```
df.drop('Salary', axis=1)
```

Handling Missing Data

```
df.dropna()
df.fillna(0)
```

Group By

```
df.groupby('Age').mean()
```

Sorting Data

```
df.sort_values('Age')
```

Merging Data

```
pd.merge(df1, df2, on='id')
```

Exporting Data

```
df.to_csv('output.csv')
```

Example

```
import pandas as pd

data = {
  'Name': ['John', 'Alice'],
  'Marks': [80, 90]
}

df = pd.DataFrame(data)
print(df)
```

Advantages of Pandas

- Easy to use
- Powerful analysis
- Handles large data

Disadvantages of Pandas

- High memory usage
- Can be slow with very large data

Best Practices

- Clean data before analysis
- Use vectorized operations
- Avoid loops